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1. Business Understanding

This section introduces the problem that will motivate the necessity of an ETL pipeline, the goals to achieve, and key stakeholders who benefit from the outcomes.

1.1 Problem Statement

The Brazilian Ministry of Education (MoE) has conducted a census of all schools in the country every year with the Censo Escolar programme. This data set includes more than 370 attributes for each school record for the period from 2010 to 2021, this including teaching staff, infrastructure, facilities, and enrollment. The raw CSV version of the data is close to 2.5GB and it is over three million rows, with all attributes of the data being stored in a single flat table, which results in considerable data redundancy like categorical data. The volume of the data and the inefficiency of the structure makes it difficult to analyze with other common tools like spreadsheets and typical database imports. Therefore, the challenges of this dataset is how can the stakeholders like policy makers, researchers, and data engineers efficiently ingest, restructure, and derive valuable insights and knowledge from this large, multi-year dataset without needing specialized infrastructure.

1.2 Business Objectives

- Apache Spark as a tool to design and implement an extract, transform and load (ETL) pipeline to ingest, clean and restructure the raw data into a scalable pipeline that supports new data.
- Transform the flat CSV structure into a Star Schema that is optimized for analytical querying.
- Load the transformed data into a relational database of PostgreSQL service available for standard SQL queries.
- Provide yearly trend analysis of important education indicators like total enrollments, numbers of teachers, and school infrastructure availability.

1.3 Stakeholders

- Ministry of Education which consolidated on figuring numbers of enrollment and staffing to inform resource allocation.
- Environmental scientists which require cleaning and structured the data to analyze patterns in access and equity.

- Data engineers tasked to keep maintaining the pipeline and ensuring that it can be reproduced.

2. Data Understanding

This section contains the source of the dataset to be used in the pipeline, including its structure, key attributes, and the relational data model designed for analytical use.

2.1 Dataset Overview

The source of the dataset is taken from Microdata of the Basic Education School Census which records detailed information about every school, student enrollments, and teacher in Brazil's basic education system. This dataset has been officially published annually by National Institute for Educational Studies and Research Anísio Teixeira (INEP) under the Brazilian Ministry of Education. Each yearly ZIP archive file in this dataset contains one large semicolon-delimited CSV file that is encoded in ISO-8859-1. All files were downloaded from Brazil's open data portal from 2010 to 2021.

The dataset contains more than 370 attributes for each census year and each of the rows represents one school. The attributes generally cover school identification and school type, administrative classification, physical infrastructure, number of teachers, and the number of students across educational level, age group, and ethnicity. In this pipeline, 15 key attributes were chosen that related to location, infrastructure indicators, enrollments, and teacher characteristics at all education levels.

Attribute	Detail
Dataset Name	Microdados do Censo Escolar da Educacao Basica
Source	Brazilian MoE open data portal
Years Covered	2010 to 2021
Raw total size	~2.5GB across 12 files
Records after load	~2,793,020 rows
Attributes per file	370+ columns

2.2 Key Attributes Selected

Column	Type	Description
nu_ano_censo	Integer	Census year from 2010-2021
no_uf	String	State name
sg_uf	String	State abbreviation
co_uf	String	State numeric code
no_municipio	String	City name
co_municipio	String	City numeric code
tp_dependencia	Integer	Admin type (1= Federal, 2= State, 3= City, 4= Private)
tp_localizacao	Integer	Location type (1= Urban, 2= Rural)
in_agua_potavel	Integer	Potable water available
in_energia_inexistente	Integer	No electricity supply
in_esgoto_inexistente	Integer	No sewage system
in_banheiro	Integer	Bathroom available
in_biblioteca	Integer	Library available
in_refeitorio	Integer	Cafeteria available
in_computador	Integer	Computer available
in_internet	Integer	Internet access available
in_equip_nenhum	Integer	No equipment at all
qt_mat_bas	Integer	Total basic education enrollments
qt_mat_inf	Integer	Enrollments for early childhood
qt_mat_fund	Integer	Enrollments for primary education
qt_mat_med	Integer	Enrollments for secondary education
qt_doc_bas	Integer	Number of teachers in basic education
qt_doc_fund	Integer	Number of teachers in primary education
qt_doc_med	Integer	Number of teachers in secondary education

2.3 Relational Data Model

The raw census data is one massive flattened table with more than 370 columns that may result in more redundancy for categorical values like state name and infrastructure indicators. To resolve this, a relational data model, using the normalization principles, including third normal form (3NF) has been designed that satisfies each non-key attribute must be related to the primary key and has no transitive attributes. All of the categorical attributes were split into their own lookup tables and monotonically increasing IDs were used with the `monotonically_increasing_id()` function in Spark. The measurements table stores the data in the center using these tables as foreign keys to show there are no repeated categorical values. The table structures, primary keys, foreign keys, and relationships between each entity are as in Figure 1.

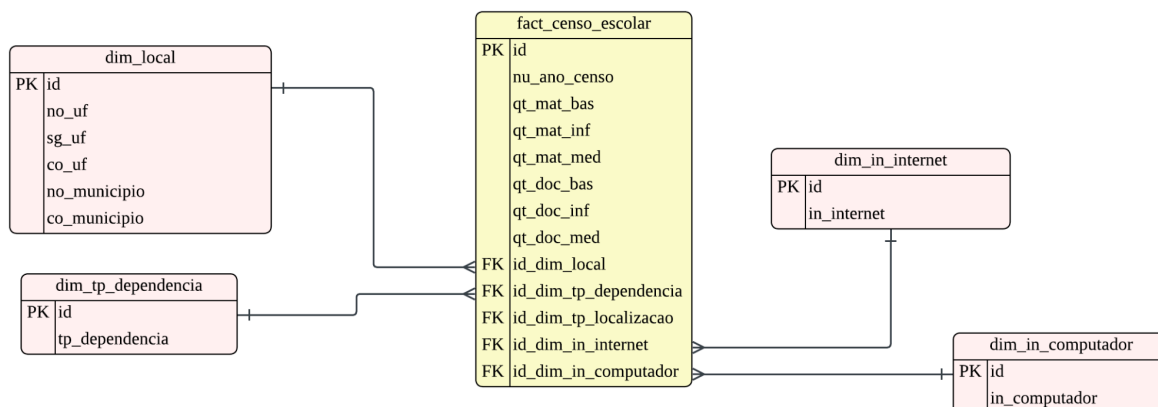


Figure 1: Entity Relationship Diagram of Censo Escolar

3. Data Preparation

The data preparation process was implemented using Apache Spark and Python to clean, transform, and prepare the CSV files before loading them into PostgreSQL. Apache Spark was used to process large datasets efficiently and support distributed data processing. The ETL workflow was automated using PySpark scripts to ensure consistency and reproducibility throughout the project.

Data Loading

The CSV files were extracted from the dataset folder and loaded into Spark DataFrames using the following configurations:

- Header enabled
- Semicolon (;) delimiter
- ISO-8859-1 encoding
- Automatic schema inference

These configurations ensured compatibility with the original dataset structure and allowed Spark to correctly identify the data types of the columns. The CSV files were loaded automatically from the dataset directory which reduced manual effort.

Data Cleaning and Transformation

Several preprocessing operations were applied before storing the data in PostgreSQL. Duplicate records were removed using `distinct()` to improve data consistency and columns were standardized using appropriate data types such as integer and string. Next, Spark casting and transformation functions were used to handle missing and inconsistent values. Then, dimension tables were created. Unique IDs were generated using `monotonically_increasing_id()` for the dimension tables. The transformation process also involved selecting relevant columns, joining dimension identifiers into the fact dataset, and preparing the final schema for database insertion.

Data Storage

The ETL pipeline stored processed data in PostgreSQL using Spark JDBC integration. The dimension tables and a fact table were created automatically through the PySpark workflow which contains primary keys and foreign keys. This ensured that all analytical records remained properly linked to their corresponding dimensions.

The entire workflow was automated using PySpark, making the ETL process repeatable, scalable, and easy to maintain. This pipeline can be executed again whenever new datasets are available.

4. Modeling

This project implemented a multidimensional data architecture using Apache Spark for ETL processing and PostgreSQL for storage. A star schema architecture was implemented to support analytical querying and improve data organization. The model consists of one central fact table, 'fact_censo_escolar', connected to multiple dimension tables such as 'dim_local', 'dim_tp_dependencia', 'dim_tp_localizacao', and infrastructure-related dimensions.

The fact table stores quantitative measures including the number of teachers and student enrollments ('QT_DOC_*', 'QT_MAT_*') together with the census year ('NU_ANO_CENSO'). Dimension tables store descriptive attributes related to school location, dependency type, internet access, bathrooms, libraries, and other facilities. Figure 2 shows the data architecture of the project whereas Figure 3 shows the star schema of the project.

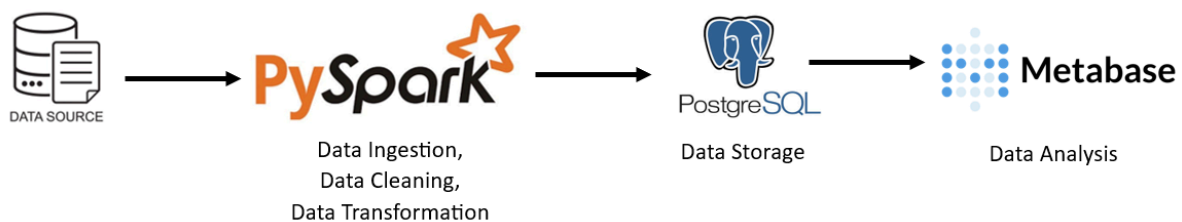


Figure 2: Data Architecture of Censo Escolar

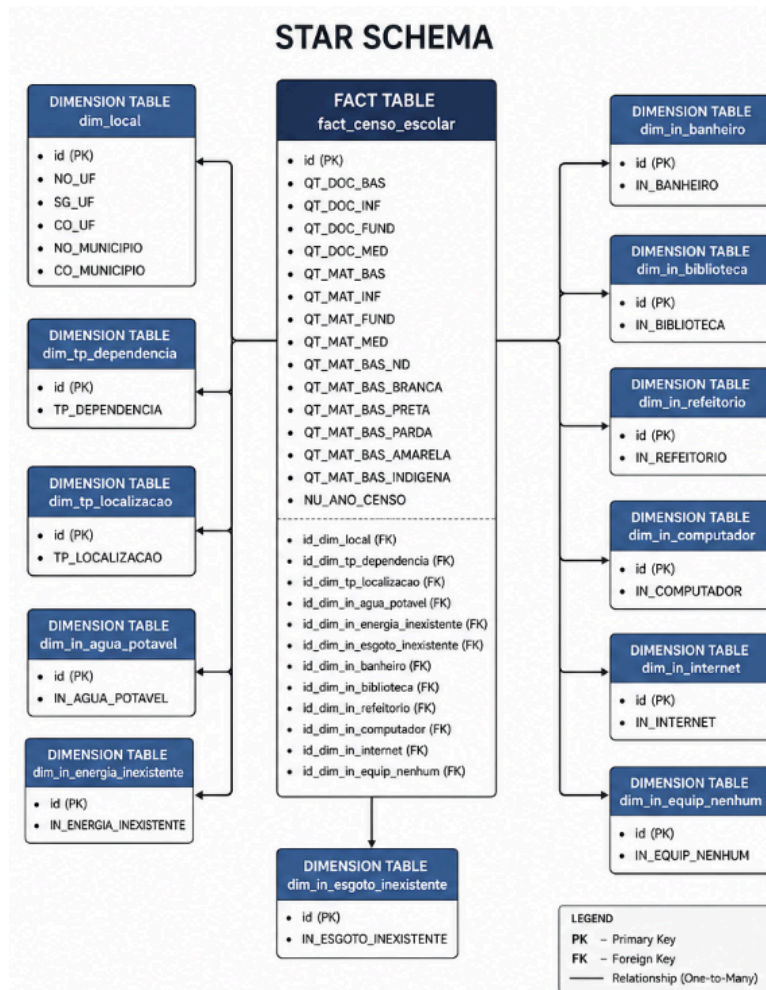


Figure 3: Star Schema of Censo Escolar

5. Evaluation

The evaluation phase focused on assessing the effectiveness of the ETL pipeline, the multidimensional data warehouse model, and the final dashboard visualization. The processed educational census data was successfully transformed into an analytical dashboard that supports reporting, trend analysis, and decision-making.

The dashboard was developed using the structured data stored in PostgreSQL and it demonstrates that the ETL pipeline successfully transformed the raw CSV files into meaningful business insights. Figure 4 shows the dashboard created using the cleaned and transformed data stored in PostgreSQL.

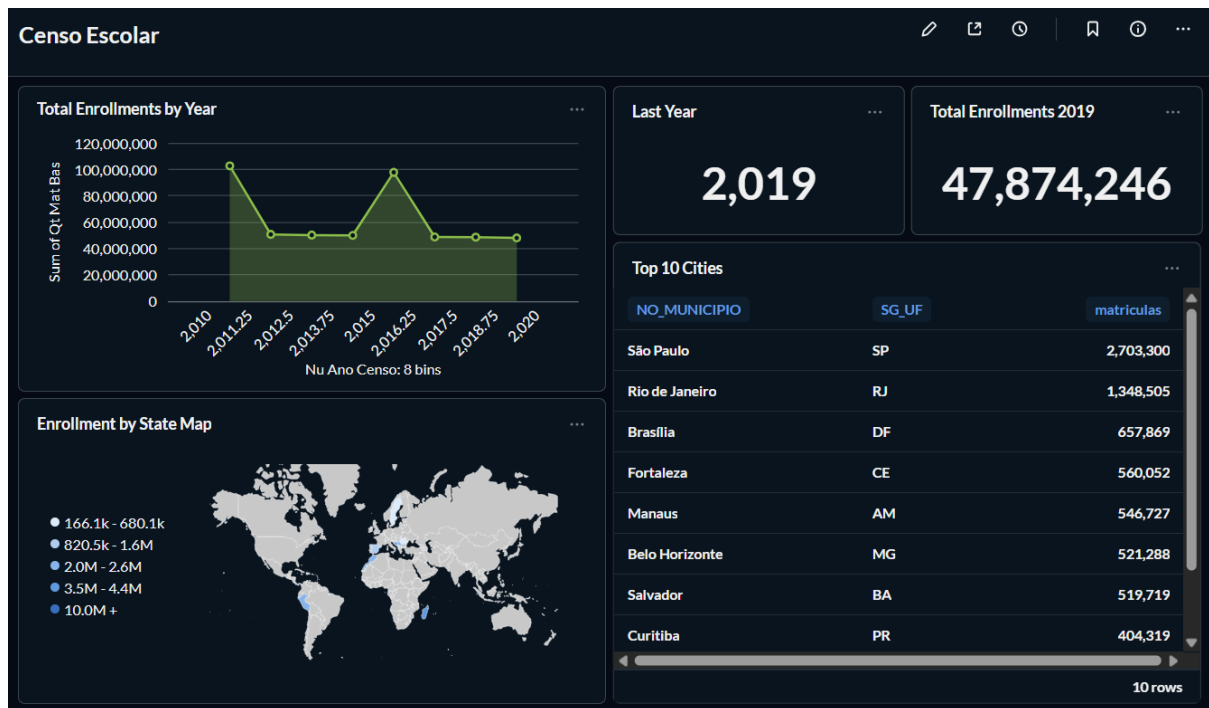


Figure 4: Dashboard of Censo Escolar

Apache Spark improved the overall data quality through standardizing data types, removing duplicate records, organizing data using structured dimensions and fact tables and by assigning primary and foreign keys. This preprocessing workflow ensured that the final dataset was structured and suitable for reporting and analysis. The star schema allows faster aggregation queries and simplifies reporting tasks, improving the overall analytical querying.

6. Reflection

- **NABIL AFLAH BOO**

This tutorial has covered all the steps of the ETL lifecycle from raw data extraction to visualization on a dashboard in Metabase. The most difficult part was to configure the Java runtime environment on Windows platform as the JAVA_HOME variable was using the older version of Java which is Java 8, while PySpark 4.1.1 required Java 17 version. Additionally, the files winutils.exe and hadoop.dll needed to be configured separately, as these are needed for Apache Spark on Windows to access the file system. This tutorial has also further educated on the concept of working with the lazy evaluation model of Spark by doing transformations that construct a query plan without executing it until a write action is performed. This process allows for efficient processing of large-scale data. Star Scheme design was also gained to break down a flat 370 columns data into 12 dimension tables and one fact table. Lastly, the

Metabase's drag and drop interface was also fascinating in showing how it is possible for data exploration without the need to write SQL.

- **HARINI**

This tutorial gave me an insight on how to process millions of data in a short period. The Censo Escolar dataset contained roughly 2.7 million data with almost 370 columns per dataset and using Apache Spark and PySpark made it easier to ingest, clean and transform these data. In addition, this project introduced me to new tools and technologies for data engineering and visualization. Previously, I had experience using Microsoft Power BI through the MS Azure tutorial, while this project allowed me to learn Docker and PostgreSQL integration as part of the ETL workflow. Through this hands-on project, I gained a better understanding of how an ETL pipeline operates in a real-world environment, from extracting raw data to transforming it into structured information suitable for dashboard visualization and analysis.